



Mycobacterium tuberculosis

- Lung Cavity size: small
- Higher chance of dissemination
- Mantoux: False negative
- ATT started first and then after 2 weeks ART is Initiated ($CD4 < 50$ cells/mcl)
- If ART is started first → Leads to **Immune Reconstitution Inflammatory Syndrome**
- **For PLHIV with TB meningitis**
 - ART should be delayed for at least 4 weeks and initiated within 8 weeks
- **For MDR/RR-TB irrespective of HIV status**
 - 1st choice - BpaLM
 - Bedaquiline: QT prolongation
 - Pretomanid: Inhibit mycolic acid synthesis
 - Linezolid
 - Moxifloxacin: QT prolongation

Mycobacterium Avium Intracellulare

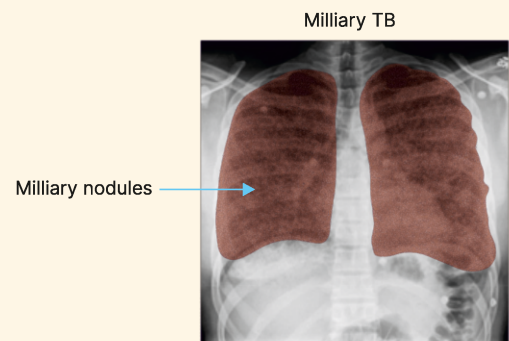
- $CD_4 < 50$ cells/mm³
- Entry via lungs or GIT
- CXR finding: Bilateral infiltrates
- IOC: Blood, Bone marrow or BAL cultures
- T/t: Clarithromycin and Ethambutol

Lymphocytic Interstitial Pneumonia

- Antibody mediated damage to lungs
- Hazy bilateral lung fields but no organism has been isolated

Important Information

- Miliary TB -
 - Low CD4 count
 - Multiple white spots seen in both lung fields
 - Classical snowstorm appearance

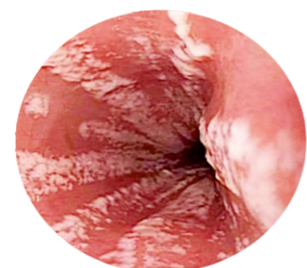


GIT DISEASES IN AIDS PATIENT

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Oesophageal candidiasis

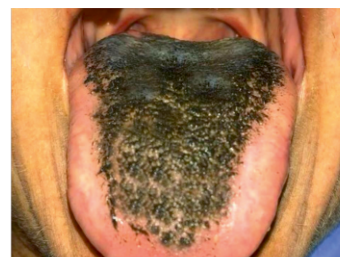
- Presents with odynophagia
- Diagnosis: UGIE
 - Curdy whitish lesions seen
- Treatment: IV Fluconazole



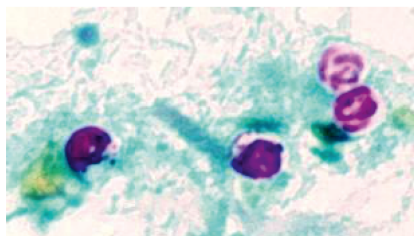


Black hairy tongue

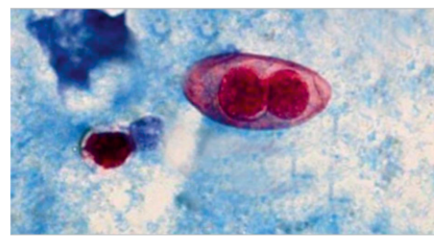
- Seen with Epstein Barr Virus



Chronic diarrhoea persisting for >1 month



Cryptosporidium



Cystoisospora

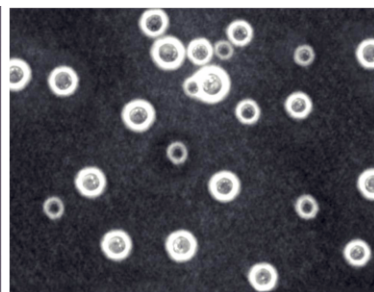
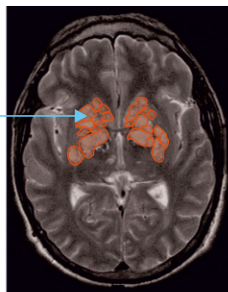
- On Kinyoun staining (AFS)
 - If cyst < 10µm → Cryptosporidium
→ Nitazoxanide (DOC)
 - If cyst > 10µm → Cystoisospora
→ Cotrimoxazole (DOC)

CNS MANIFESTATIONS IN AIDS

00.49.06

Cryptococcal meningitis

Soap bubble appearance



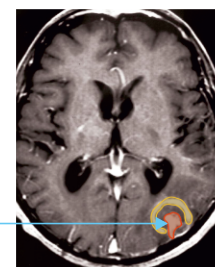
India ink preparation

- IOC: CSF ELISA for Cryptococcal antigen > Lateral flow assay
- India ink preparation (not done) – can be false negative d/t low fungal count
- MRI head finding: Soap bubble appearance
- T/t: Liposomal Amphotericin B + 5-Flucytosine

Cerebral toxoplasmosis

- Clinical presentation: AIDS positive patient with recurrent episodes of Seizures
- Hyponatremia
- MRI finding: Eccentric target sign
- T/t: DOC - Pyrimethamine + Sulfadiazine + Leucovorin

Eccentric target sign

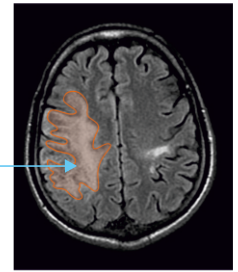




Progressive Multifocal Leukoencephalopathy

- Caused by JC virus
- Neurodegeneration involving white matter subsequently grey matter
- MRI finding: Subcortical hyperintensity
- Treatment: ART

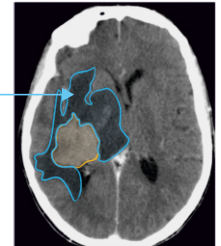
Subcortical
hyperintensity



Primary CNS lymphoma

- Focal neurological deficits + seizures
- Diagnosis: Stereotactic brain biopsy

Lesion with
perilesional
edema



Important Information

- Causes of seizures in HIV+ patients: HIV encephalopathy > Cerebral toxoplasmosis > Primary CNS lymphoma
- M/C type of lymphoma in HIV positive patient – Immunoblastic lymphoma (Subtype of Diffuse large B Cell lymphoma)

TREATMENT (COMBINATION FORMULATIONS FOR AIDS)

00:55:41

- **Tenofovir + Lamivudine + Dolutegravir** - Both therapeutic and post exposure prophylaxis
- If the patient cannot take integrase inhibitor → Efavirenz + Emtricitabine + Tenofovir Disoproxil Fumarate (Longest clinical experience)
- Dolutegravir + Abacavir + Lamivudine
- Dolutegravir + Emtricitabine + Tenofovir Alafenamide (TAF)
 - Tenofovir – effective against hepatitis B infection

ACCIDENTAL NEEDLE STICK INJURY

00:57:17

- Chance of transmission: **0.3 %**
- Post exposure prophylaxis
 - Started ASAP (< 1 hr)
 - Not effective after 72 hrs
 - Duration: 28 days
 - T/t: Tenofovir + Lamivudine + Dolutegravir



Q. Match the following regarding tests used in HIV infection:

1. Most sensitive test for HIV	a. Nucleic Acid Amplification Test
2. Most specific test for HIV	b. PCR-HIV DNA
3. Best test for early diagnosis of HIV infection	c. Enzyme immunoassay
4. Best test for vertical transmission of HIV	d. Western Blot

- a. 1-c, 2-d, 3-a, 4-b
- b. 1-a, 2-c, 3-b, 4-d
- c. 1-d, 2-b, 3-c, 4-a
- d. 1-c, 2-a, 3-b, 4-d

Ans. (a)



TRACERS CoV-2/COVID-19

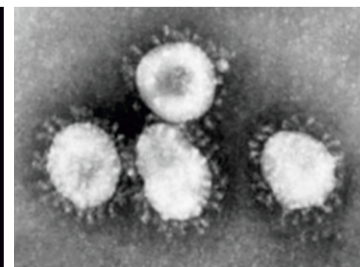
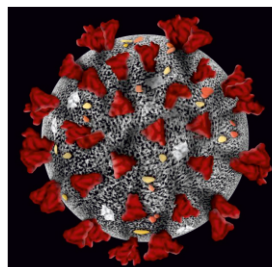


- Leading cause of death in COVID-19 → **ARDS**

INTRODUCTION

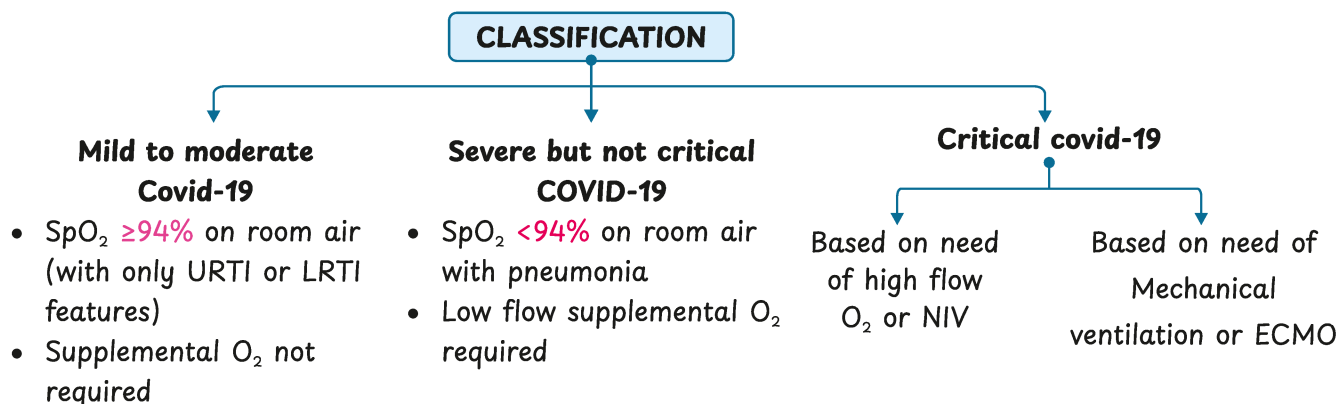
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- Belongs to the **Coronaviridae** family
- Single-stranded +RNA virus with a large genome of 30 Kb
 - Replication errors can occur in virus → Delta strain (High mortality rate) of the virus converted into the Omicron strain (Low mortality rate)
- Reservoir: Bat
- Ro: **5-6** (For Influenza A → Ro: 1-2)
- Virus (Spike antigen hiding behind the petal shaped peplomers) binds to human ACE2 protein receptor in lungs → Replication of virus starts
- Advanced age is the principal risk factor : ↑ mortality
- M/c complication
 - **ARDS** (non-cardiogenic pulmonary edema)
 - Thrombo-embolism
- Incubation period: 4-5 days
- Virus is detectable up to 3 weeks - months



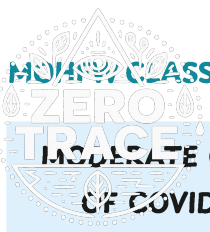
ASSESSMENT OF CLINICAL SEVERITY TO TARGET TREATMENTS

00:07:07



RISK FACTORS

- Age >60 yrs
- BMI >25
- DM
- HTN
- CVD
- Chronic lung disease
- Cancer or immunocompromised patients



MODERATE CASES OF COVID-19

- SpO₂ <94% on room air
- RR >24/min
- Interventions
 - Non-rebreather mask with O₂ flow rate @ 15 L/min
 - Dexamethasone/ Methylprednisolone (↓ Mortality rates)
 - Enoxaparin SC (For microthrombi → D/t ventilation-perfusion mismatch)

SEVERE CASES OF COVID-19

- SpO₂ <90% on room air
- RR >30/min
- Interventions
 - Low work of breathing
 - NIV
 - High flow nasal cannula with O₂ flow rate @ 60 L/min
 - High work of breathing
 - Tracheostomy (or)
 - Mechanical ventilation (Low volume ventilation with low tidal volume)
 - Dexamethasone/ Methylprednisolone
 - Enoxaparin SC

RECOMMENDED INTERVENTIONS ACCORDING TO IDSA (INFECTIOUS DISEASES SOCIETY OF AMERICA)

00:15:08

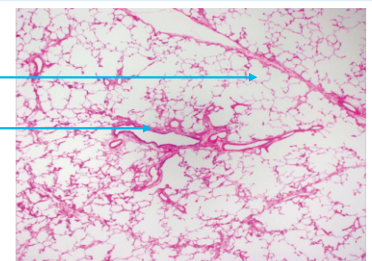
- Ambulatory mild to moderate patients with a high risk of progression to severe disease of hospitalisation
 - T/t
 - Nirmatrelvir + Ritonavir : For adults and children >12 yrs of age (should start with 5 days of symptom onset)
 - Molnupiravir: For adults with risk factors (should start with 5 days of symptom onset)
 - Remdesivir: For adults and children (give IV infusions at a health care facility for 3 days)
- Hospitalized severe to critical COVID-19 pneumonia patients
 - T/t
 - Corticosteroids for 10 days (Dexamethasone, Methylprednisolone)
 - IV Remdesivir
 - Tocilizumab (IL-6 receptor blocker) or Sarilumab in progressive disease with elevated inflammatory markers (TLC, IL-6)
 - Baricitinib or Tofacitinib (JAK-STAT signalling pathway inhibitors) in patients with elevated inflammatory markers

PATHOPHYSIOLOGY

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NORMAL LUNG PARENCHYMA

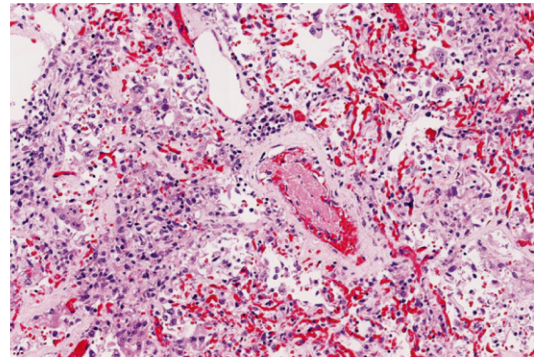
Normal alveoli
Normal blood vessel



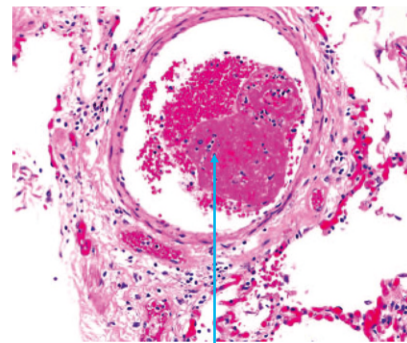
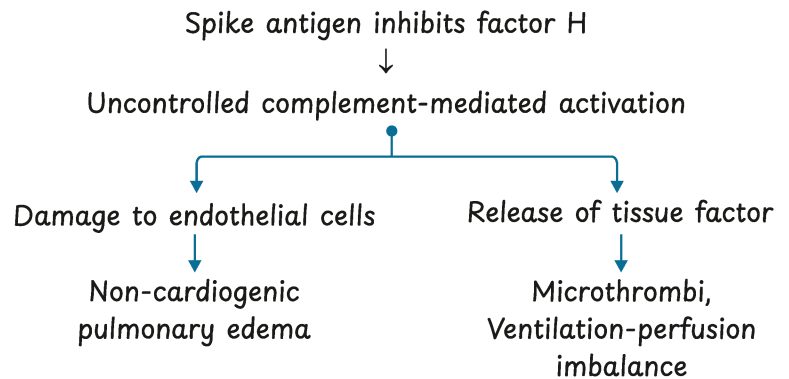


IMPACT OF COVID-19 ON LUNGS

- **Thickened alveolar septa**
- Inflammation
- Endothelial cell injury → Non cardiogenic pulmonary edema



MICROTHROMBI



Microthrombi

Multidimensional Challenge of Treating Covid-19

Stage/ Severity:	Asymptomatic/ Presymptomatic	Mild Illness	Moderate Illness	Severe Illness	Critical Illness
	SARS-COV2 test positive but no symptoms	Mild symptoms (eg, fever, cough, taste/smell changes); no dyspnea	O ₂ saturation ≥ 94%, lower respiratory tract disease	O ₂ saturation < 94%, respiratory rate >30/min; lung infiltrate >50%	Respiratory failure, shock, multi organ dysfunction/failure
Frequency:	-	80	80	15%	5%



Disease pathogenesis:

Viral replication

Inflammation

Potential treatment:

Antivirals

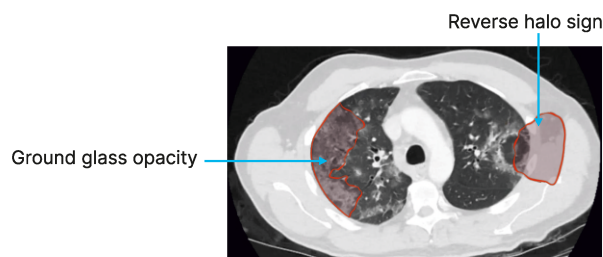
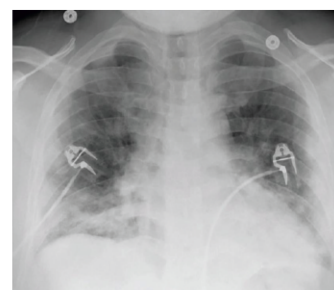
Antibody therapy

Decrease Inflammation

WORKUP

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- **RT-PCR (IOC)**
 - Sample is taken using nasal swabs with the patient's neck extended to 70° and depth of insertion → 8-10 cm
 - Wooden swabs should not be used
- Other test to be done in severe illness patients with COVID-19
 - ↓ TLC, Leukopenia
 - Neutrophil-Lymphocyte ratio >6 (Higher the ratio, poorer the prognosis)
 - D-dimer assay: ↑
 - CRP, LDH, IL-6, Ferritin: ↑
 - Troponin I, Pro-BNP
- Chest X-RAY- **Multiple bilateral opacities** in the middle and lower-lung zones, towards the periphery
- HRCT chest findings:
 - Peripheral Ground glass opacities
 - Crazy pavement pattern
 - **Reverse Halo sign**



STEP UP OXYGEN DELIVERY FOR CRITICAL COVID-19 PATIENTS

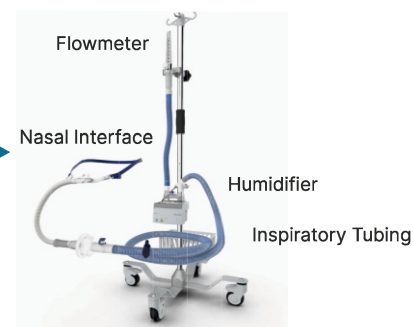
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Tight fitting mask



Helmet interface

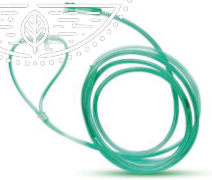


Non-rebreather mask (@15L/min)

→ NIV

High flow nasal cannula

→ Mechanical ventilation



1. NC @ 6LPM



+



3. NC + Non -rebreather



5. NIPPV: CPAP



2. Venturi mask up to 50%

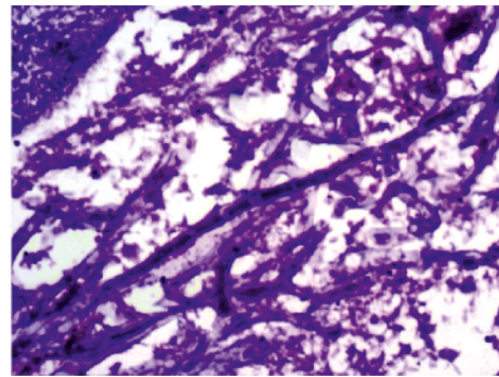
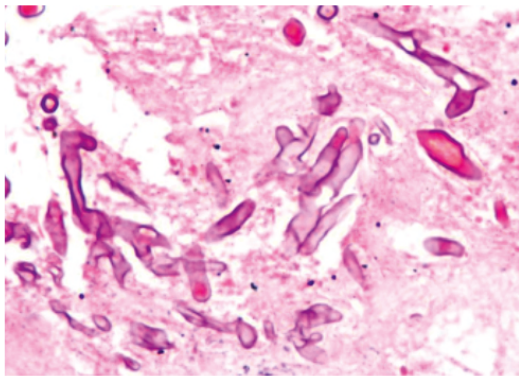


4. HFNC



6. Intubation

COMPLICATIONS POST COVID-19



Important Information

- Black fungus- Dematiaceous fungi



Q. Which of the following is not a part of the lung protective strategy in a critically sick patient with COVID-19?

- a. Prone position
- b. Low tidal volume
- c. Low PEEP
- d. Dexamethasone

Ans. (c)

Q. Best for the diagnosis of cheek swelling & loosening of teeth in patient treated for COVID-19?

- a. MRI
- b. Nasal swab and culture
- c. β -D-Glucan test
- d. Blood culture

Ans. (a)